makes sense if certain conditions are satisfied. The second formulation that we discuss is called information filtering, derived in Section 6.2. Information filtering propagates the inverse of the covariance matrix (i.e., P^{-1}) instead of P, and is computationally cheaper than Kalman filtering under certain conditions. The third formulation that we discuss is called square root filtering, derived in Section 6.3. Square root filtering effectively increases the precision of the Kalman filter, which can help prevent divergence and instability. However, this is at the cost of increased computational effort. The final formulation that we discuss is called U-D filtering, derived in Section 6.4. This is another way to implement square root filtering, which helps to prevent numerical difficulties in the implementation of the Kalman filter.

6.1 SEQUENTIAL KALMAN FILTERING

In this section, we derive the sequential Kalman filter. This is a way of implementing the Kalman filter without matrix inversion. This can be a great advantage, especially in an embedded system that may not have matrix routines. However, the use of sequential Kalman filtering only makes sense if certain conditions are satisfied, which we will discuss in this section.

Recall the Kalman filter measurement update formulas from Equation (5.16):

$$\begin{aligned}
y_k &= H_k x_k + v_k \\
K_k &= P_k^- H_k^T (H_k P_k^- H_k^T + R_k)^{-1} \\
\hat{x}_k^+ &= \hat{x}_k^- + K_k (y_k - H_k \hat{x}_k^-) \\
P_k^+ &= (I - K_k H_k) P_k^-
\end{aligned} \tag{6.1}$$

The computation of K_k requires the inversion of an $r \times r$ matrix, where r is the number of measurements. This is depicted in Figure 6.1.

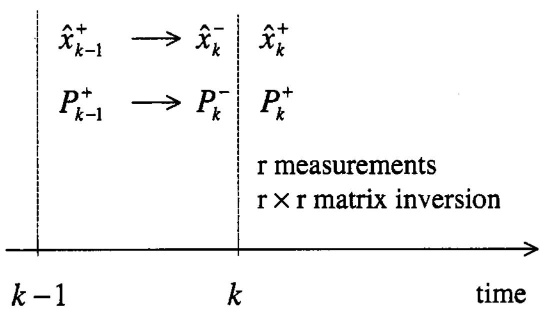

Figure 6.1 The measurement-update equation of the standard Kalman filter requires an $r \times r$ matrix inversion, where r is the number of measurements.

Suppose that instead of measuring y_k at time k, we obtain r separate measurements at time k. That is, we first measure $y_k(1)$, then $y_k(2)$, $\cdots$, and finally $y_k(r)$. We will use the shorthand notation y_{ik} for the ith element of the measurement vector y_k. Assume for now that R_k (the covariance of measurement y_k) is diagonal;

that is, R_k is given as

$$R_k = \begin{bmatrix} R_{1k} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & R_{rk} \end{bmatrix} \tag{6.2}$$

We will also use the notation that H_{ik} is the ith row of H_k, and v_{ik} is the ith element of v_k. Then we obtain

$$\begin{aligned} y_{ik} &= H_{ik} x_k + v_{ik} \\ v_{ik} &\sim (0, R_{ik}) \end{aligned} \tag{6.3}$$

So instead of processing the measurements at time k as a vector, we will implement the Kalman filter measurement-update equation one measurement at a time. We use the notation that K_{ik} is the Kalman gain that is used to process the ith measurement at time k, $\hat{x}_{ik}^+$ is the optimal estimate after the ith measurement has been processed at time k, and P_{ik}^+ is the estimation-error covariance after the ith measurement at time k has been processed. We can see from these definitions that

$$\begin{aligned} \hat{x}_{0k}^+ &= \hat{x}_k^- \\ P_{0k}^+ &= P_k^- \end{aligned} \tag{6.4}$$

That is, $\hat{x}_{0k}^+$ is the estimate after zero measurements have been processed, so it is equal to the *a priori* estimate. Similarly, P_{0k}^+ is the estimation-error covariance after zero measurements have been processed, so it is equal to the *a priori* estimation-error covariance. The gain K_{ik} and covariance P_{ik}^+ are obtained from the normal Kalman filter measurement-update equations, with the understanding that they apply to the scalar measurement y_{ik}. For $i = 1, \cdots, r$ we have

$$\begin{aligned} K_{ik} &= P_{i-1,k}^+ H_{ik}^T (H_{ik} P_{i-1,k}^+ H_{ik}^T + R_{ik})^{-1} \\ \hat{x}_{ik}^+ &= \hat{x}_{i-1,k}^+ + K_{ik}(y_{ik} - H_{ik}\hat{x}_{i-1,k}^+) \\ P_{ik}^+ &= (I - K_{ik} H_{ik}) P_{i-1,k}^+ \end{aligned} \tag{6.5}$$

After all r measurements are processed, we set $\hat{x}_k^+ = \hat{x}_{rk}^+$, and $P_k^+ = P_{rk}^+$, and we have our *a posteriori* estimate and error covariance at time k. The sequential Kalman filter does not require any matrix inversions because all of the expressions in Equation (6.5) are scalar operations. This process is depicted in Figure 6.2. The sequential Kalman filter can be summarized as follows.

The sequential Kalman filter

1. The system and measurement equations are given as

$$\begin{aligned} x_k &= F_{k-1} x_{k-1} + G_{k-1} u_{k-1} + w_{k-1} \\ y_k &= H_k x_k + v_k \\ w_k &\sim (0, Q_k) \\ v_k &\sim (0, R_k) \end{aligned} \tag{6.6}$$

   where w_k and v_k are uncorrelated white noise sequences. The measurement covariance R_k is a diagonal matrix given as

$$R_k = \operatorname{diag}(R_{1k}, \cdots, R_{rk}) \tag{6.7}$$

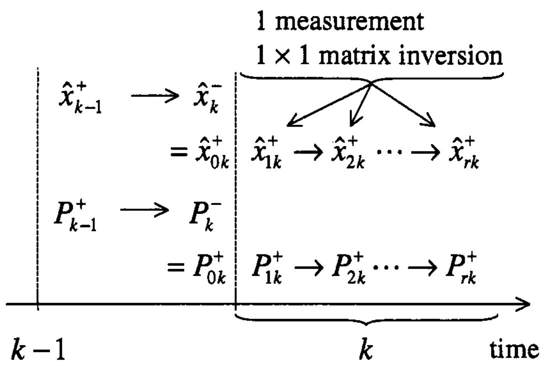

Figure 6.2 The measurement update equation of the sequential Kalman filter requires r scalar divisions (where r is the number of measurements) because the measurements at each time step are processed sequentially. This is in contrast to the standard Kalman filter processing that is depicted in Figure 6.1.

2. The filter is initialized as

$$\begin{aligned} \hat{x}_0^+ &= E(x_0) \\ P_0^+ &= E[(x_0 - \hat{x}_0^+)(x_0 - \hat{x}_0^+)^T] \end{aligned} \tag{6.8}$$

3. At each time step k, the time-update equations are given as

$$\begin{aligned} P_k^- &= F_{k-1}P_{k-1}^+F_{k-1}^T + Q_{k-1} \\ \hat{x}_k^- &= F_{k-1}\hat{x}_{k-1}^+ + G_{k-1}u_{k-1} \end{aligned} \tag{6.9}$$

   This is the same as the standard Kalman filter.

4. At each time step k, the measurement-update equations are given as follows.

   (a) Initialize the *a posteriori* estimate and covariance as

$$\begin{aligned} \hat{x}_{0k}^+ &= \hat{x}_k^- \\ P_{0k}^+ &= P_k^- \end{aligned} \tag{6.10}$$

   These are the *a posteriori* estimate and covariance at time k after zero measurements have been processed; that is, they are equal to the *a priori* estimate and covariance.

   (b) For $i = 1, \cdots, r$ (where r is the number of measurements), perform the following:

$$\begin{aligned} K_{ik} &= \frac{P_{i-1,k}^+H_{ik}^T}{H_{ik}P_{i-1,k}^+H_{ik}^T + R_{ik}} \\ &= \frac{P_{ik}^+H_{ik}^T}{R_{ik}} \\ \hat{x}_{ik}^+ &= \hat{x}_{i-1,k}^+ + K_{ik}(y_{ik} - H_{ik}\hat{x}_{i-1,k}^+) \\ P_{ik}^+ &= (I - K_{ik}H_{ik})P_{i-1,k}^+(I - K_{ik}H_{ik})^T + K_{ik}R_{ik}K_{ik}^T \end{aligned}$$

$$\begin{aligned} &= \left[(P_{i-1,k}^{+})^{-1} + H_{ik}^{T} H_{ik}/R_{ik}\right]^{-1} \\ &= (I - K_{ik}H_{ik})P_{i-1,k}^{+} \end{aligned} \tag{6.11}$$

(c) Assign the *a posteriori* estimate and covariance as

$$\begin{aligned} \hat{x}_k^{+} &= \hat{x}_{rk}^{+} \\ P_k^{+} &= P_{rk}^{+} \end{aligned} \tag{6.12}$$

The development above assumes that the measurement-noise covariance R_k is diagonal. What if R_k is not diagonal? Suppose that $R_k = R$ is not diagonal, but it is a constant matrix. We perform a Jordan form decomposition of R by finding a matrix S such that

$$R = S\hat{R}S^{-1} \tag{6.13}$$

$\hat{R}$ is a diagonal matrix containing the eigenvalues of R, and S is an orthogonal matrix (i.e., $S^{-1} = S^T$) containing the eigenvectors of R. This decomposition is always possible if R is symmetric positive definite, as discussed in most linear systems books [Bay99, Che99, Kai00]. Now define a new measurement $\tilde{y}_k$ as

$$\begin{aligned} \tilde{y}_k &= S^{-1}y_k \\ &= S^{-1}(H_k x_k + v_k) \\ &= \tilde{H}_k x_k + \tilde{v}_k \end{aligned} \tag{6.14}$$

where $\tilde{H}_k$ and $\tilde{v}_k$ are defined by the above equation. The covariance of $\tilde{v}_k$ can be obtained as

$$\begin{aligned} E(\tilde{v}_k\tilde{v}_k^T) &= E(S^{-1}v_k v_k^T S^{-T}) \\ &= E(S^{-1}v_k v_k^T S) \\ &= S^{-1}E(v_k v_k^T)S \\ &= S^{-1}RS \\ &= \hat{R} \end{aligned} \tag{6.15}$$

So we have introduced a normalized measurement $\tilde{y}_k$ that has a diagonal noise covariance. Now we can implement the sequential Kalman filter equations, except that we use the measurement $\tilde{y}_k$ instead of y_k, the measurement matrix $\tilde{H}_k$ instead of H_k, and the measurement noise covariance $\hat{R}$.

Note that this procedure would not make sense if R were time-varying, because in that case we would have to perform a Jordan form decomposition at each step of the Kalman filter. That would be a lot of computational effort in order to avoid a matrix inversion. However, if R is constant and it is known before the implementation of the Kalman filter, then we can perform the Jordan form decomposition offline and use the sequential Kalman filter to our advantage.

In summary, it only makes sense to use the sequential Kalman filter if one of the following two conditions holds:

1. The measurement noise covariance R_k is diagonal
2. The measurement noise covariance R is a constant.

Finally, note that the term sequential filtering is sometimes used synonymously with the Kalman filter. That is, sequential is often used as a synonym for recursive [Buc68, Chapter 13], [Bro96]. This can cause some confusion in terminology. However, sequential filtering is usually used in the literature as we use it in this section; that is, sequential filtering is a filtering method that processes measurements one at a time (rather than processing the measurements as a whole vector). Sometimes, the standard Kalman filter is called the batch Kalman filter to distinguish it from the sequential Kalman filter.

EXAMPLE 6.1

The change x_k from one week to the next of an American football team's ranking is related to the team's performance against that week's opponent. The expected relationships between various normalized game measures y_{ik} and the team's ranking change at the kth week are given as

$$\begin{aligned} y_{1k} &= x_k + v_{1k} = \text{ point differential} \\ y_{2k} &= \frac{1}{5}x_k + v_{2k} = \text{ turnover differential} \\ y_{3k} &= \frac{1}{50}x_k + v_{3k} = \text{ yardage differential} \end{aligned} \tag{6.16}$$

where $v_{1k} \sim (0,2)$, $v_{2k} \sim (0,1)$, and $v_{3k} \sim (0,50)$. Before the first game of the season is played, it is expected that the team ranking will increase by one due to certain players having returned from injuries. The variance of this *a priori* estimate is 4. Uncertainty in ownership conditions is expected to decrease the team's ranking by 5% each week, with a variance of 2. The system can therefore be modeled as

$$\begin{aligned} x_{k+1} &= 0.95x_k + w_k \\ y_k &= \begin{bmatrix} 1 & 1/5 & 1/50 \end{bmatrix}^T x_k + v_k \\ w_k &\sim (0,Q) \quad Q = 2 \\ v_k &\sim (0,R) \quad R = \text{diag}(2,1,50) \\ \hat{x}_0^+ &= 1 \\ P_0^+ &= 4 \end{aligned} \tag{6.17}$$

Suppose that the team plays its first game and wins by six points, gains three more turnovers than its opponent, and is outgained by 100 yards. That is, $y_1 = \begin{bmatrix} 6 & 3 & -100 \end{bmatrix}^T$. The standard Kalman filter adjusts the team's ranking as follows:

$$\begin{aligned} P_1^- &= FP_0^+F^T + Q \\ &= 5.61 \\ \hat{x}_1^- &= 0.95\hat{x}_0^+ \\ &= 0.95 \\ K_1 &= P_1^-H^T(HP_1^-H^T + R)^{-1} \\ &= \begin{bmatrix} 0.6961 & 0.2785 & 0.0006 \end{bmatrix} \\ \hat{x}_1^+ &= \hat{x}_1^- + K_1(y_1 - H\hat{x}_1^-) \end{aligned}$$

$$
\begin{aligned}
&= 5.1922 \\
P_1^+ &= (I - K_1 H)P_1^- \\
&= 1.3923
\end{aligned} \tag{6.18}
$$

The K_1 calculation requires the inversion of a 3×3 matrix. On the other hand, the sequential Kalman filter could be used to update the estimated team ranking as follows:

$$
\begin{aligned}
P_1^- &= F P_0^+ F^T + Q \\
&= 5.61 \\
\hat{x}_1^- &= 0.95 \hat{x}_0^+ \\
&= 0.95 \\
\hat{x}_{01}^+ &= \hat{x}_1^- \\
P_{01}^+ &= P_1^-
\end{aligned} \tag{6.19}
$$

The first measurement is processed as follows:

$$
\begin{aligned}
K_{11} &= P_{01}^+ H_1^T (H_1 P_{01}^+ H_1^T + R_{11})^{-1} \\
&= 0.7372 \\
\hat{x}_{11}^+ &= \hat{x}_{01}^+ + K_{11}(y_{11} - H_1 \hat{x}_{01}^+) \\
&= 4.6728 \\
P_{11}^+ &= (I - K_{11} H_1) P_{01}^+ \\
&= 1.4744
\end{aligned} \tag{6.20}
$$

The second measurement is processed as follows:

$$
\begin{aligned}
K_{21} &= P_{11}^+ H_2^T (H_2 P_{11}^+ H_2^T + R_{22})^{-1} \\
&= 0.2785 \\
\hat{x}_{21}^+ &= \hat{x}_{11}^+ + K_{21}(y_{21} - H_2 \hat{x}_{11}^+) \\
&= 5.2479 \\
P_{21}^+ &= (I - K_{21} H_2) P_{11}^+ \\
&= 1.3923
\end{aligned} \tag{6.21}
$$

The third measurement is processed as follows:

$$
\begin{aligned}
K_{31} &= P_{21}^+ H_3^T (H_3 P_{21}^+ H_3^T + R_{33})^{-1} \\
&= 0.0006 \\
\hat{x}_{31}^+ &= \hat{x}_{21}^+ + K_{31}(y_{33} - H_3 \hat{x}_{21}^+) \\
&= 5.1922 \\
P_{31}^+ &= (I - K_{31} H_3) P_{21}^+ \\
&= 1.3923
\end{aligned} \tag{6.22}
$$

The sequential Kalman filter requires three loops through the measurement update equations, but no matrix inversions are required.

$\nabla\nabla\nabla$